

LESSON 9-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Equations from Tables**

Maryland Standard 6.EE.C.9 • Learning target – I can write an equation that matches a table of values.

REMEMBER

- An **equation** like $y = 5x$ is a rule that works for **every** row.
- Same $y \div x$ in every row: $y = (\text{that number}) \times x$. Same $y - x$: $y = x + (\text{that number})$.
- Check your equation with **two** rows.

WORKED EXAMPLEWrite an equation for x : 2, 3, 5 and y : 12, 18, 30.**Step 1** $12 \div 2 = 6$, $18 \div 3 = 6$, $30 \div 5 = 6$ **Step 2** Check: $6 \times 5 = 30$ **Answer** $y = 6x$ **WORD BANK****equation** – a rule with an equal sign, like $y = 5x$ **check** – test your rule on another row**rate** – the amount for each 1**PRACTICE***show your work • label units*

1. Write an equation for each table.

Step 1 Divide each y by its x . Same number every time? Write $y = (\text{number})x$.**Step 2** If not, find $y - x$. Same number? Write $y = x + (\text{number})$.**a.** x : 1, 2, 3 y : 7, 14, 21 $y =$ _____**b.** x : 1, 2, 3 y : 10, 11, 12 $y =$ _____**c.** x : 2, 4, 6 y : 10, 20, 30 $y =$ _____**d.** x : 5, 6, 7 y : 2, 3, 4 $y =$ _____**e.** x : 0, 4, 8 y : 6, 10, 14 $y =$ _____**f.** x : 1, 3, 5 y : 9, 27, 45 $y =$ _____**g.** x : 2, 3, 4 y : 16, 24, 32 $y =$ _____**h.** x : 10, 12, 14 y : 5, 7, 9 $y =$ _____

2. Write the equation. Then complete the table.

Step 1 Divide 22 by 2 to find the rate.**Step 2** Use the rate to fill in the empty boxes.

x	2	3	4	6	9
y	22	33	44		

Equation _____

- 3.
- Stretch.**
- A table shows
- x
- : 2, 5, 9 and
- y
- : 0, 3, 7. Write an equation. Then find
- y
- when
- $x = 20$
- .

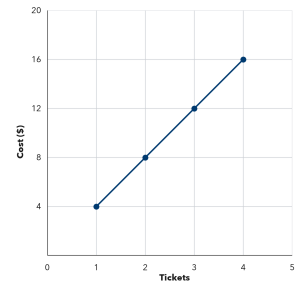
Answer _____

- 4. Tickets.** The graph shows the cost of fair tickets. Write an equation for the graph. Then find the cost of 10 tickets.

Step 1 Read one point on the graph, like (1, 4).

Step 2 Divide the cost by the tickets. Then multiply by 10.

Answer _____



- 5. Find the Error.** For x : 2, 4, 6 and y : 6, 8, 10, Owen wrote $y = 3x$ because $3 \times 2 = 6$. Explain his mistake and write the correct equation.

Frame Owen's equation only works for _____. The correct equation is $y = \underline{\hspace{1cm}}$.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Make a 3-row table for a rule. Ask someone at home to find your equation.

LESSON 9-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Equations from Situations**

Maryland Standard 6.EE.C.9 • Learning target – I can write an equation from a described situation and identify its variables.

REMEMBER

- A **rate** ("each," "per") multiplies x . A **starting amount** or fee is added once.
- $y = (\text{rate}) \times x + (\text{starting amount})$. Check it with one value from the story.

WORKED EXAMPLE

A gym charges \$10 to enter plus \$7 per hour. Write an equation for the cost y of x hours.

Step 1 Rate: 7 per hour, so $7x$. Fee: add 10.

Step 2 Check 3 hours: $7 \times 3 + 10 = 31$

Answer $y = 7x + 10$

WORD BANK

rate – "each" or "per"; it multiplies x
fee – a cost paid one time; it is added

starting amount – what you have at the start; it is added

PRACTICE

show your work • label units

- Write an equation for each situation.
 - Each pack holds 12 pencils. Pencils y in x packs. $y = \underline{\hspace{2cm}}$
 - Jamal is 3 years older than his sister. Jamal's age y when she is x . $y = \underline{\hspace{2cm}}$
 - A club costs \$20 to join plus \$15 a month. Cost y for x months. $y = \underline{\hspace{2cm}}$
- Bowling.** Bowling shoes cost **\$4** one time. Each game costs **\$5**. Write an equation for the cost y of x games. Find the cost of 4 games.

Step 1 Multiply the games by \$5.

Step 2 Add the \$4 shoe fee one time.

Answer _____
- Taxi.** A taxi charges \$3 to start plus \$2 per mile. Write an equation for the cost y of x miles. Find the cost of 7 miles.

Step 1 Multiply the miles by \$2.

Step 2 Add the \$3 start fee one time.

Answer _____
- Car Wash.** A class has \$30 and earns \$8 per car washed. Complete the table. Write an equation.

Step 1 At $x = 0$, no cars are washed yet. What does the class have?

Step 2 Add \$8 for each car.

Cars (x)	0	1	2	5
Total \$ (y)				

Equation _____

- 5. Find the Error.** For “a \$25 fee plus \$10 an hour,” Nia wrote $y = 35x$. Explain her mistake and fix it.

Frame Nia’s mistake is _____. The fee is paid _____, so $y =$ _____.

TEST PRACTICE

choose the best answer

- 6.** A plumber charges a \$40 fee plus \$25 per hour. Which equation gives the cost c for h hours?

A $c = 40h + 25$

B $c = 25h + 40$

C $c = 65h$

D $c = 25h$

- 7.** Which situation matches $y = 3x + 12$?

A 3 cans per box, 12 boxes

B Start with \$3, earn \$12 per hour

C Start with 12 gal, add 3 gal per min

D \$12 for 3 people

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____